

THE TAPE STREET v1.2.8 PROTOCOL

Technical Specification & Security Architecture Provider: Big Widget LLC

Classification: Proprietary / High-Entropy Deployment

I. EXECUTIVE SUMMARY

The Tape Street v1.2.8 "Scorched Earth" protocol represents a paradigm shift in local data hardening. By moving beyond traditional software-based encryption and anchoring the cryptographic process to physical silicon, Big Widget LLC has eliminated the common vectors of "portable" brute-force attacks.

II. THE "SCORCHED EARTH" ENGINE (ARGON2ID)

At the core of v1.2.8 is a highly tuned implementation of the Argon2id Key Derivation Function (KDF).

- **Memory-Hard Constraint:** The protocol enforces a **128MB RAM bottleneck**.
- **Handshake Latency:** Optimized for a **0.63s** user response time on native x64 architecture.
- **Attack Mitigation:** This RAM requirement forces an attacker to dedicate massive hardware resources per password guess, rendering GPU and ASIC-based parallel attacks mathematically unfeasible.

III. HARDWARE-BOUND ENTROPY (SILICON ANCHORING)

Unlike standard encryption, v1.2.8 does not rely on a standalone password. The "Master Key" is a composite of user input and unique hardware identifiers:

1. **CPU Signature:** Unique processor ID and instruction set mapping.
2. **Motherboard UUID:** Permanent hardware serial identification.
3. **The Result:** The resulting ciphertext is "locked" to the physical machine. If the ZIP or EXE is moved to secondary hardware, the internal "Salt" changes, and decryption becomes mathematically impossible even with the correct password.

IV. ANTI-FORENSIC & TEMPORAL DEFENSES

To combat modern automated cracking suites, v1.2.8 employs two primary defensive layers:

- **Temporal Lockout:** A forced **60-second hardware gate** is triggered upon failed attempts, limiting potential brute-force speed to 1,440 attempts per 24 hours.

- **Zero-Trace Execution:** Developed on the Python 3.14 AST and compiled via the Zig C++ backend, the binary utilizes native memory management to minimize the "RAM footprint" of sensitive keys post-execution.

V. MATHEMATICAL ORIGINS: THE $3n+1$ FOUNDATION

The entropy logic of Tape Street is inspired by the complexity of the Collatz Conjecture ($3n+1$). By utilizing non-linear mathematical progression, the protocol ensures that the relationship between the input (password) and the output (encrypted data) is as unpredictable as the hailstone sequences of the $3n+1$ problem, providing a 415-million-year brute-force timeline against current compute standards.

VI. DEPLOYMENT SPECIFICATIONS

- **Runtime:** Native Windows x64 (Standalone)
 - **Entropy Source:** User String + Hardware ID Forge
 - **Encryption Standard:** AES-256-GCM / Argon2id
 - **Status:** Commercial Grade / Field Ready
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